Standards-Compliant Laser Transmitter Design, Wavelength Control and Nonlinearity Management for an 8-Channel DWDM Link

ECA1804 – Optoelectronics and Optical Communication
CO2 PROBLEM SOLVING ASSIGNMENT

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Assumptions used where data was not given:

D = 17 ps/(nm·km) at 1550 nm (so $|\beta_2| \approx 21.7$ ps²/km, consistent with the given 20 ps²/km); g_R = 6×10^{-14} m/W and g_B = 5×10^{-11} m/W (the Raman exponent was lost in the PDF; 6×10^{-14} is the standard silica value); Brillouin linewidth $\Delta\nu_B \approx 20$ MHz; EDFA noise figure 5 dB; diode voltage ≈ 1.5 V; Class 1 eye-safety limit at 1550 nm ≈ 10 mW (10 dBm) CW

Stage 1 – Interpret the Communication Requirement

Link: 8 × 25 Gb/s = 200 Gb/s on a 100 GHz grid (8 × 0.8 nm = 6.4 nm occupied, e.g. 193.1–193.8 THz = 1552.52–1546.92 nm). Path: 6 × 80 km = 480 km of SMF, amplified every span. Span loss = 80 × 0.2 = 16 dB.

Why the source must be fast, narrow-line and powerful

- Speed: NRZ at 25 Gb/s needs B ≈ 0.7 R_b = 17.5 GHz; the bit period is only 40 ps.
- Narrow linewidth/high SMSR: each channel's NRZ main lobe is about 50 GHz wide (null-to-null ≈ 2R_b), so a 100 GHz slot leaves little room. Side modes or a wide spectrum cause inter-channel crosstalk and, over 480 km, a large dispersion penalty ($\Delta T = D\cdot L\cdot\Delta\lambda$).
- Power: span loss of 16 dB and six cascaded EDFAs set the OSNR. OSNR (0.1 nm) ≈ 58 + P_ch − L_span − NF − 10 log N = 58 + 3.0 − 16 − 5 − 7.8 ≈ **32 dB**, which leaves roughly 8–10 dB margin over the ≈ 22 dB typically needed by 25G NRZ (assumed).

Source parameters that fix wavelength, efficiency and spectral purity:

bandgap ($\lambda = 1.24/Eg$), internal quantum efficiency $\eta int = \tau_{nr}/(\tau r + \tau nr)$, threshold current Ith, slope efficiency ηs, linewidth (set by spontaneous emission and the α-factor), side-mode suppression ratio (SMSR).

Quantities to control for reliable high-speed operation:

relaxation-oscillation frequency fr, 3 dB modulation bandwidth, α-factor (linewidth enhancement) and chirp, extinction ratio, and temperature (wavelength drift 0.1 nm/°C, threshold shift).

Nonlinear quantities that matter:

Leff, γP·Leff (SPM), channel spacing and dispersion (XPM, FWM phase matching), per-channel power (SBS) and total power (SRS), with all of these accumulating over 6 spans.

Standards:

- ITU-T G.957 (with G.691/G.698/G.694.1 for DWDM): mean launched power window, centre wavelength/grid, spectral width, SMSR ≥ 30 dB, extinction ratio.
- IEEE 802.3bm/bs (25G NRZ and 50G PAM4 lanes of 100/200/400GbE): per-lane launch power/OMA, wavelength, SMSR, extinction ratio, transmitter dispersion penalty (TDP/TDECQ), RIN and eye mask.
- ANSI Z136.1/Z136.2: the accessible optical power at 1550 nm is capped, and an open-fibre hazard needs automatic laser shutdown (ALS) and labelling.

System parameters for soliton/DSP and integration: β_2, γ, pulse width T_0, launch power, number of channels, receiver type (direct vs coherent), footprint/power/cost.

Problem Requirement Statement

The transmitter must deliver eight single-longitudinal-mode channels on a 100 GHz grid near 1550 nm, each carrying 25 Gb/s NRZ with ≥ 17.5 GHz usable bandwidth. Each channel needs ≈ 2 mW launch power (≥ 3 dBm) and SMSR ≥ 35 dB. Wavelength must stay within ±0.1 nm over a 40 °C ambient swing. The chirp must be low enough for 480 km. Nonlinear penalties must be controlled (SBS, SPM, XPM, FWM, SRS). The design must meet G.957/IEEE 802.3bm/bs-type transmitter specs and eye-safety limits, and be realisable as a compact, scalable, reliable module.

Stage 2 – Optical Source Selection

Recombination and bandgap. Carriers injected into the active region recombine radiatively (useful photons, τ_r = 10 ns) and non-radiatively (defects, Auger, surface; τ_nr = 40 ns). The bandgap sets the photon energy.

- Wavelength: $\lambda \approx 1.24/Eg = 1.24/0.80 =$ **1.55 µm = 1550 nm ✔** (centre of the C-band).
- Total lifetime: $1/\tau = 1/10 + 1/40 = 0.125\ ns^{-1}$, so τ = 8 ns.
- Internal quantum efficiency: $\eta int = \tau/\tau r = \tau nr/(\tau r + \tau nr) = 40/(10 + 40) =$ **0.80 (80 %)**.
- Differential efficiency check: $\eta d = \eta_s \cdot q/(h\nu) = 0.3$ W/A ÷ 0.80 V = 0.375. So $\eta d/\eta int \approx$ 0.47: about half of the internally generated photons are extracted (mirror loss vs internal loss).

Threshold condition. Oscillation starts when round-trip gain equals round-trip loss: $R_1R_2 \cdot \exp[2(\Gamma gth - \alpha_int)L] = 1$, i.e. $\Gamma g_th = \alpha_int + (1/2L) \cdot \ln(1/R_1R_2)$, with phase condition $2\beta L = 2\pi m$. Above I_th, stimulated emission clamps the carrier density, narrows the line and gives high-speed, high-power output. Below I_th (LED regime) the output is broad and slow. A narrow linewidth plus SMSR ≥ 35 dB matters in an 8-channel link because side modes from 7 neighbours add as crosstalk (7 interferers at −35 dB ≈ −26.5 dB) and because chirped/wide spectra walk off in dispersion.

Comparison of candidates (typical values at 1550 nm)

Parameter	LED	FP	DBR	DFB	VCSEL (1550)	QD laser
Output power (fibre)	≈ 0.01–0.1 mW	1–10 mW	5–20 mW	5–20 mW	1–3 mW	2–10 mW
Spectral width	50–100 nm	1–5 nm (multimode)	< 10 MHz, SLM	1–5 MHz, SLM	SLM, few–10s MHz	broad/comb or SLM
SMSR	n/a	none	> 35 dB	> 40 dB	> 30 dB	design-dependent
Modulation speed	≤ 0.2 GHz	~10 GHz	10 GHz	10–30 GHz	10–25 GHz	10–25 GHz
Efficiency / threshold	low, no threshold	moderate, 10–30 mA	moderate, mode-hop risk	good, 5–15 mA	very low threshold (1–3 mA)	very low, high T_0
Fibre coupling	poor (broad beam)	fair	fair	good (with lens/EML)	excellent (circular beam)	fair
8-ch DWDM suitability	✘	✘	✔ (tunable)	✔✔	limited (power, 1550 maturity)	emerging

Design Decision 1

I recommend a 1550 nm InGaAsP/InP MQW DFB laser (CW, coupled to an external modulator) optical source because: (1) $\lambda = 1.24/0.80 = 1550$ nm sits on the DWDM grid and in the minimum-loss window; (2) $\eta_int = 80$ % and $\eta_d = 37.5$ %, giving 8.4 mW at 40 mA (Stage 3); (3) a single longitudinal mode with SMSR > 40 dB and MHz-level linewidth suits 100 GHz channels and 480 km; (4) its 20 GHz bandwidth exceeds the 17.5 GHz requirement, and (5) it is mature, thermally predictable (0.1 nm/°C) and integrates with EAMs. LEDs and FP lasers fail on spectral width and speed.

Credible alternative: a DBR laser becomes preferable if one tunable part must cover all eight channels (sparing/reconfigurable network), accepting a more complex multi-section control and mode-hop risk. A directly modulated DFB becomes preferable for short, unamplified metro links ($\lesssim$ 10 km) where cost and power dominate.

Stage 3 – Modulation, Chirp and Spectral Stability

Output power. $P = \eta_s(I - I_{th}) = 0.3$ mW/mA × (40 − 12) mA = **8.4 mW**; $P(\text{dBm}) = 10 \log_{10}(8.4)$ = **+9.24 dBm**.

Bandwidth. $B \approx 0.7\, R_b = 0.7 \times 25$ = **17.5 GHz**. Available 20 GHz, so the margin = 2.5 GHz (14.3 %, ratio 1.14 ≈ 0.6 dB). Sufficient but thin, so the laser should not be pushed to the edge: an external modulator with ≥ 25 GHz bandwidth is preferred.

Rate equations (N = carrier density, S = photon density, $G = v_g \cdot a \cdot (N - N_{tr})$):

- $dN/dt = I/(qV) - N/\tau - G \cdot S$
- $dS/dt = \Gamma \cdot G \cdot S - S/\tau_p + \Gamma \cdot \beta_{sp} \cdot N/\tau_r$

The relaxation-oscillation frequency is $f_r = (1/2\pi) \cdot \sqrt{(\Gamma \cdot v_g \cdot a \cdot S_0/\tau_p)} \propto \sqrt{(I - I_{th})}$ (the $I/I_{th} - 1$ factor here is 40/12 − 1 = 2.33). For well-damped lasers $f_{3dB} \approx 1.55\, f_r$, so $f_r \approx 20/1.55 \approx 13$ GHz.

- Small-signal: linearise about the bias (S_0, N_0); the response is a second-order low-pass with a peak at f_r. The bandwidth is set by f_r and the damping rate.
- Large-signal (NRZ on/off): the laser is driven through threshold, so turn-on delay and relaxation ringing appear, causing pattern-dependent jitter and overshoot, and large transient chirp. Keeping the low level above threshold reduces this.

Direct vs external modulation

Aspect	Direct (DML)	External (EAM/MZM)
Chirp	$\alpha \approx 3$–5; $\Delta\nu \approx (\alpha/4\pi)(1/P)(dP/dt)$ + adiabatic term	$\alpha \approx 0$ to ±0.7, near transform-limited
Bandwidth	limited by f_r (≈ 20 GHz)	EAM/MZM 30–50 GHz
Dispersion interaction	chirp + D → rapid broadening	much smaller broadening
Reach at 25G NRZ	≈ 2–10 km	tens–hundreds of km with compensation
Cost/loss	lowest, no insertion loss	4–8 dB insertion loss, extra driver

Order-of-magnitude chirp for a DML with $\alpha = 4$: $\Delta\nu \approx (4/4\pi) \times (1/14\text{ ps}) \approx 20$ GHz ≈ 0.15 nm. After one 80 km span, $D \cdot L = 17 \times 80 = 1360$ ps/nm, so $\Delta T = 1360 \times 0.15 \approx 200$ ps ≈ 5 bit periods (40 ps). Over 480 km: $D \cdot L = 8160$ ps/nm. Direct modulation fails for this link.

Temperature. Uncontrolled drift = 0.1 nm/°C × 40 °C = **4 nm = 5 channel spacings** (4/0.8), so the channel would collide with neighbours. For ±0.1 nm: $\Delta T \leq 0.1/0.1$ = **±1 °C** minimum; the design uses a TEC at **±0.1 °C (≈ ±0.01 nm)** to leave budget for ageing and locker error.

TEC vs injection-current tuning

- TEC: large, slow (≈ s), large range (several nm), no effect on the modulation point. Costs TEC power.
- Injection current: fast but small range (≈ 0.003–0.01 nm/mA, mostly self-heating), and it changes output power, extinction ratio and chirp. Suitable only for trimming, not for compensating a 4 nm drift.

Design Decision 2

I recommend external (EAM-integrated, EML) modulation and TEC-based wavelength control with a wavelength locker (etalon + photodiode feedback, trimmed by heater/current) because: the laser gives 8.4 mW (+9.2 dBm) of CW power, the modulator's ≥ 25 GHz bandwidth exceeds the 17.5 GHz need with margin, chirp is near zero (avoiding ≈ 200 ps/span broadening), and a TEC at ±0.1 °C keeps drift to ±0.01 nm against a 4 nm uncontrolled swing and a ±0.1 nm tolerance. Dispersion modules (DCF or FBG) in each span handle the remaining 8160 ps/nm.

Stage 4 – Nonlinearities

Effective length. $\alpha = 0.2$ dB/km ÷ 4.343 = 0.04605 km^{-1}; $\alpha L = 0.04605 \times 80 = 3.684$; e^(−3.684) = 0.0251. L_eff = (1 − 0.0251)/0.04605 = **21.2 km**.

SPM. $\varphi_{NL} = \gamma \cdot P \cdot L_{eff} = 1.3$ W^{-1}km^{-1} × 0.002 W × 21.17 km = **0.055 rad per span**, so **0.33 rad over six spans** (upper bound, phase adds each span). This is well below the ≈ 1 rad level where SPM-dispersion penalty becomes significant. Pessimistic bound including all 8 channels in the same phase ($P_{total} = 16$ mW): $\gamma \cdot P_{tot} \cdot L_{eff} = 0.44$ rad/span ≈ 2.6 rad over six spans. In practice XPM is far smaller because of walk-off (below).

XPM and FWM.

- XPM: a neighbour's intensity changes a channel's phase (up to $2\gamma P_{other} \cdot L_{eff}$). With dispersion, channels walk off: 0.8 nm × 17 ps/nm/km = 13.6 ps/km, so a 40 ps bit slips past in ≈ 3 km, averaging XPM out. XPM converts to intensity noise and timing jitter through dispersion. It grows with launch power and with closer spacing/low dispersion.
- FWM: three channels generate $f_{ijk} = f_i + f_j - f_k$, which fall on grid slots at 100 GHz spacing, giving in-band crosstalk. Phase mismatch $\Delta\beta = \beta_2(2\pi\Delta f)^2 = 21.7$ ps²/km × (2π × 100 GHz)² ≈ **8.6 km^{-1}**, so $L_{coh} = 2\pi/\Delta\beta \approx 0.74$ km ≪ L_{eff}. FWM efficiency $\eta \approx \alpha^2/(\alpha^2 + \Delta\beta^2) = 0.0021/73.1 \approx 2.9\times10^{-5}$ (−45 dB). Then $P_{FWM} \approx \eta \cdot (6)^2 \cdot \gamma^2 \cdot P^3 \cdot L_{eff}^2 \approx 6$ nW ≈ **−52 dBm** per product per span (≈ −55 dB relative to 2 mW); even worst-case coherent accumulation over 6 spans (×36) gives about −39 dB. FWM is small here because dispersion is high (non-zero-dispersion fibre or DSF would make it much worse).

SBS and SRS thresholds

- SBS: $P_{th} = 21 \cdot A_{eff}/(g_B \cdot L_{eff}) = 21 \times 80\times10^{-12}$ m² ÷ (5×10^{-11} m/W × 21.17×10^{3} m) = $1.68\times10^{-9} \div 1.0585\times10^{-6}$ = **1.59 mW (+2.0 dBm)**.

- SRS: $P_{th} = 16 \cdot A_{eff}/(g_R \cdot L_{eff}) = 16 \times 80\times10^{-12} \div (6\times10^{-14} \times 21.17\times10^{3}) = 1.28\times10^{-9} \div 1.27\times10^{-9} \approx$ **1.0 W (+30 dBm)**.

Check	Power	vs threshold
2 mW per channel vs SBS	2 mW	1.26× **above** 1.59 mW (+1.0 dB) ✘
2 mW per channel vs SRS	2 mW	504× below (−27 dB) ✔
16 mW total vs SRS	16 mW	63× below (−18 dB) ✔

Important finding: an unmodulated narrow-linewidth source at 2 mW exceeds the SBS threshold, so it must be actively mitigated. Raman tilt across the 5.6 nm band is small: power transfer $\approx g_R \cdot P_{tot} \cdot L_{eff}/A_{eff} \times$ (700 GHz/13.2 THz) $\approx 0.25 \times 0.053 \approx 1.3$ %, about 0.1 dB end-to-end per span (< 1 dB over six spans), correctable with gain equalisation.

SBS vs SRS

	SBS	SRS
Origin	photons scattered by acoustic phonons	scattering by optical phonons (molecular vibrations)
Direction	backward only	forward (and backward)
Gain bandwidth	≈ 20 MHz, Stokes shift ≈ 11 GHz	≈ 13 THz wide, peak at ≈ 13 THz (~100 nm)
Power transfer	to a back-travelling Stokes wave of the same channel; saturates forward power	from shorter-λ to longer-λ channels (tilt/crosstalk)
Effect on DWDM	per-channel limit (channels 100 GHz apart do not interact) → noise/limited launch power	total-power and bandwidth-dependent; matters for many wide-spaced channels
Threshold	≈ 1.6 mW (low)	≈ 1 W (high)

Source linewidth: if the optical linewidth is $\Delta\nu_s$, the effective SBS threshold scales roughly as $P_{th}(1 + \Delta\nu_s/\Delta\nu_B)$. With $\Delta\nu_B \approx 20$ MHz and a ≈ 200 MHz dither/broadened spectrum: $1.59 \times 11 \approx$ **17.5 mW**, i.e. 9.4 dB above 2 mW. A very narrow-linewidth laser is therefore worse for SBS.

Eye safety (ANSI Z136). Assumed Class 1 limit at 1550 nm ≈ 10 mW (10 dBm). Total launch 16 mW = **12.0 dBm**, i.e. 1.6× (+2 dB) over that limit. A single channel (2 mW = 3 dBm) is safely below it. So the booster output port requires Class 1M-type handling (ITU-T G.664 hazard-level logic), with ALS/automatic power reduction and labelling. If strict Class 1 is

required, cap the total at 10 mW (1.25 mW/ch, also below the SBS threshold); the 2 dB loss is covered by the ≈ 8–10 dB OSNR margin.

Two practical mitigation methods without cutting received signal level:

1. Use large-effective-area fibre ($A_{eff} \approx 125\ \mu m^2$ gives $\gamma \approx 1.3 \times 80/125 \approx 0.83\ W^{-1}km^{-1}$, −36 % phase) and keep dispersion non-zero.
2. Use hybrid distributed-Raman/EDFA amplification (lower effective noise figure and flatter power profile), so the same OSNR is obtained at lower launch power. Combine with optimum span launch-power setting and SBS dither (≈ 200 MHz low-frequency phase/current dither, or accept modulated-source broadening).

Design Decision 3

I recommend an SBS-dither + dispersion-managed (large-A_{eff} fibre, per-span DCM, gain equalised EDFA/Raman) nonlinear-control strategy with ALS/Class 1M eye-safety control because: SPM is only 0.055 rad/span (0.33 rad total) and FWM ≈ −52 dBm, so these are controlled by dispersion; SRS threshold (1 W) is 63× above the total power; but SBS (1.59 mW) is exceeded by 2 mW, so dither raises it to ≈ 17.5 mW; and total power of 12 dBm needs ALS/hazard control (fallback: 1.25 mW/ch).

Stage 5 – Soliton, Digital Backpropagation and Integration

Soliton balance. In the anomalous-dispersion regime ($\beta_2 < 0$), dispersion produces a chirp that Kerr SPM (positive nonlinear chirp) cancels exactly, so the pulse keeps its sech shape when $L_D = L_{NL}$ ($N = 1$).

- $L_D = T_0^2/|\beta_2| = (15\ ps)^2/20\ ps^2/km = 225/20 =$ **11.25 km**.
- Fundamental soliton peak power ($L_{NL} = L_D$): $P_0 = 1/(\gamma \cdot L_D) = 1/(1.3 \times 11.25) =$ **0.0684 W ≈ 68 mW** per channel.
- At 2 mW: $L_{NL} = 1/(\gamma P) = 1/(1.3 \times 0.002) =$ **385 km**.
- Soliton order: $N^2 = L_D/L_{NL} = 11.25/384.6 = 0.0293$, so **$N = 0.17 \ll 1$**.

Feasibility: not feasible. At 2 mW, dispersion acts ≈ 34× faster than the Kerr effect; the pulse just spreads. A true soliton needs ≈ 68 mW per channel (≈ 0.55 W for 8 channels), which would violate eye safety and the SBS threshold. Additionally: pulse spacing 40 ps = 2.7 T_0 gives strong soliton interactions (need $\gtrsim$ 7–10 T_0); the amplifier spacing 80 km $\gg$ the soliton period $z_0 = (\pi/2)L_D = 17.7$ km; and 8-channel WDM adds collision-induced jitter. Soliton transmission is therefore rejected.

Digital backpropagation (DBP). The received (coherent) optical field is numerically propagated backwards through a virtual fibre with the opposite sign of dispersion and nonlinearity, using split-step Fourier steps (D^{-1}, then $e^{(-j\gamma P_{eff} \cdot L_{step})}$) for each span, so deterministic CD and Kerr SPM are removed. Limitation: it needs a coherent receiver with phase information and high ADC/DSP complexity (many steps per span, oversampling), and it cannot remove XPM/FWM from channels it does not process, nor ASE–signal interaction or PMD. Single-channel DBP gains only ~0.5–1.5 dB; full-field DBP requires all 8 channels' fields jointly.

Is DBP appropriate for the full 8-channel link? No. The baseline link is 25G NRZ direct-detection (no phase information), the channels are generated by 8 separate transmitters, and XPM/FWM from the other 7 channels cannot be backpropagated. DBP is only suitable as an optional coherent upgrade (sub-channel DBP). The baseline uses optical dispersion compensation instead.

Standards parameters and checks

Parameter	ITU-T G.957 (and G.691/G.698)	IEEE 802.3bm/bs	How checked
Centre wavelength / grid	window or ITU grid ±tolerance	lane wavelength range	OSA / wavemeter
Mean launch power	max/min per interface	per-lane avg. power, OMA	power meter, OMA
Spectral width / SMSR	RMS or −20 dB width, SMSR ≥ 30 dB	SMSR ≥ 30 dB	OSA
Extinction ratio	min. (≈ 8.2 dB, SDH)	min. ER	sampling scope
Eye mask / jitter	mask of pulse shape	eye mask, TDP/TDECQ	eye diagram + BERT
RIN / return loss	RIN limit, reflection tolerance	RIN, return loss	RIN meter

The prototype is verified with a PRBS-31 pattern, the eye/BERT test (BER < 10^{-12}), OSA scans over 0–40 °C, and an SBS back-reflection test.

Integration comparison

Aspect	Discrete	Monolithic InP PIC	Si-photonics + hybrid InP laser
Footprint	largest (8 TOSAs + mux)	small (8 EML + AWG on one chip)	smallest/scalable, heterogeneous
Power	8 TECs, 8 drivers	single TEC, lower	low drive, but laser still InP; heaters for tuning
Coupling	many alignments (loss, cost)	one chip-fibre coupling	grating/edge couplers, laser-to-Si loss

Aspect	Discrete	Monolithic InP PIC	Si-photonics + hybrid InP laser
Reliability	many joints	fewer joints; mature InP	bonding/yield maturity lower
Scalability	poor	good (8 → 16/32 ch.)	excellent (CMOS volume)
Source quality	best individual tuning	gain, DFB+EAM together	needs hybrid laser / external light

Integration improves system performance by reducing coupling loss and reflections (less SBS/crosstalk, better OSNR), enabling common thermal control and accurate relative alignment of the 100 GHz grid, and lowering power/cost per Gb/s. The penalty is lower yield and thermal crosstalk between neighbouring lasers.

Design Decision 4

I recommend a monolithic InP photonic-integrated transmitter (8 DFB lasers + 8 EAMs + integrated AWG mux + monitor PDs, on a single TEC) followed by a booster EDFA, with optical dispersion compensation as the baseline and DBP kept only as a future coherent option, because: it delivers chirp-free external modulation from the same source family selected in Stage 2, keeps the 100 GHz grid under one thermal loop (±0.1 °C), minimises coupling losses and back-reflections, and meets G.957/802.3bm-type parameter checks. Soliton transmission (N = 0.17) and DBP (direct detection, 8 channels) are rejected. Si-photonics with a hybrid laser is the preferred alternative at high volume or for > 16 channels.

Stage 6 – Verification Table

Requirement	Specified / Target	Achieved	Satisfactory?
Optical source	~1550 nm, SLM, narrow line	λ = 1550 nm, η_int = 80 %, DFB, SMSR > 40 dB	Yes
Modulation bandwidth and output power	≥ 17.5 GHz; ≥ 2 mW launch	20 GHz (margin 2.5 GHz, EAM ≥ 25 GHz); 8.4 mW (+9.24 dBm) before modulator	Yes
Spectral/chirp and wavelength stability	±0.1 nm over 40 °C	uncontrolled 4 nm; TEC ±0.1 °C → ±0.01 nm; EAM chirp ≈ 0	Yes

Requirement	Specified / Target	Achieved	Satisfactory?
Nonlinear control and eye safety	φ_NL small; below SBS/SRS; ≤ Class limit	φ_NL = 0.055 rad/span, 0.33 rad total; SBS 1.59 mW (exceeded by 2 mW → dither to ≈ 17.5 mW); SRS 1 W; 16 mW = 12 dBm > 10 mW → ALS/Class 1M or 1.25 mW/ch	Yes, with mitigation
Soliton / dispersion condition	N ≈ 1 for solitons	L_D = 11.25 km; P_0 = 68 mW; L_NL = 385 km; N = 0.17	Soliton rejected; dispersion managed with DCM
Integration / DSP / standards	G.957, 802.3bm/bs; scalable	monolithic InP PIC; DBP not appropriate for the 8-ch direct-detect link; checks by OSA/eye/BERT	Yes

Consistency: the calculations agree with the requirement. The two points that the data forces us to address are that 2 mW is above the 1.59 mW SBS threshold, and that 16 mW total exceeds a 10 mW Class 1 level. Both are solved by design (dither, ALS/Class 1M or a 1.25 mW/ch fallback), supported by the ≈ 32 dB OSNR (≈ 8–10 dB margin).

Final Transmitter Design – Integrated Engineering Solution

Parameter table

Parameter	Value
Source	1550 nm InGaAsP/InP MQW DFB (CW), η_int = 0.80
Channel plan	8 ch, 193.1–193.8 THz (1552.52–1546.92 nm), 100 GHz spacing
Bias / threshold	I = 40 mA, I_th = 12 mA, η_s = 0.3 mW/mA
CW output	8.4 mW (+9.24 dBm)
Modulation	EAM (integrated EML), 25 Gb/s NRZ, bandwidth ≥ 25 GHz

Parameter	Value
Laser bandwidth / margin	20 GHz vs 17.5 GHz needed (2.5 GHz)
TEC setting	25 °C set-point, ±0.1 °C (drift ±0.01 nm); wavelength locker and heater trim
Launch power	2 mW (+3 dBm) per channel, 16 mW (+12 dBm) total
SBS control	≈ 200 MHz dither → threshold ≈ 17.5 mW
Dispersion	per-span DCM, residual ≲ ±50 ps/nm (assumed)
Safety	ALS, Class 1M label (or 1.25 mW/ch for Class 1)

How the choices work together: the DFB supplies narrow-line, high-power light at the right wavelength. The EAM removes the chirp that would make a 480 km link impossible, and the TEC/locker keeps eight channels on the 100 GHz grid. Dither, large-A_eff fibre and dispersion management keep nonlinearity below penalty level, and ALS keeps the 12 dBm output safe. The InP PIC lets all of this fit in one thermally controlled package. Solitons (N = 0.17) and DBP (direct detection, 8 channels) are not used.

Standards: ITU-T G.957/G.691/G.698/G.694.1, IEEE 802.3bm/bs, ANSI Z136.1/Z136.2 (with IEC 60825-1, ITU-T G.664).

Final Engineering Decision

YES – I would deploy the transmitter solution.

Justification: the DFB at 1550 nm (η_int 80 %, 8.4 mW) meets the wavelength and power needs; the 20 GHz bandwidth exceeds the 17.5 GHz requirement; external modulation with TEC control removes the 200 ps/span chirp-dispersion penalty and holds drift at ±0.01 nm instead of 4 nm. Nonlinear effects are acceptable (φ_NL = 0.33 rad, FWM ≈ −52 dBm, SRS threshold 1 W), the one real problem (SBS at 1.59 mW) is fixed with dither, and the eye-safety issue (12 dBm vs ≈ 10 dBm) with ALS/Class 1M or a 1.25 mW/ch fallback, with OSNR ≈ 32 dB leaving margin. Solitons are infeasible (N = 0.17, 68 mW needed) and DBP is not suitable for this direct-detection 8-channel link, so dispersion compensation plus the integrated InP PIC is the realistic solution. **Deployment is conditional on prototype confirmation** of SMSR, TDP/eye mask and BER over 0–40 °C, and on the SBS and ALS tests.